

Name: _____

Date: _____

STRAIGHT-LINE GRAPHS, (EQUATION OF A STRAIGHT LINE)

LO: I can identify and write the equation of a straight line in the form $y = mx + c$.

Equation of a straight line

Every straight line can be written as $y = mx + c$, where m is the gradient (slope) and c is the y-intercept.

Reading from a graph: The y-intercept is where the line crosses the y-axis. The gradient is the steepness, calculated as $\text{rise} \div \text{run}$.

Quick examples

●	$y = 2x + 3$ has gradient 2 and y-intercept 3	$m = 2, c = 3$
●	$y = -x + 5$ has gradient -1	line slopes downward
●	$y = \frac{1}{2}x$ has gradient $\frac{1}{2}$ and y-intercept 0	passes through origin
●	$y = 3x + 2$ has gradient 2	gradient is 3, not 2

Key idea

In $y = mx + c$, m is the gradient (how steep) and c is where the line crosses the y-axis.

$$y = \textcircled{m}x + \textcircled{c}$$

$m = \text{gradient}, c = \text{y-intercept}$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - identifying m and c

State the gradient and y-intercept of $y = 3x - 2$.

Compare with $y = mx + c$
 $m = 3$ (gradient)
 $c = -2$ (y-intercept)

Gradient = 3, y-intercept = -2

Read m and c directly from the equation.

Example 2 - writing the equation

A line has gradient 4 and y-intercept -1. Write its equation.

Use $y = mx + c$
 Substitute $m = 4$ and $c = -1$
 $y = 4x - 1$

$y = 4x - 1$

Plug the gradient and y-intercept into the formula.

Example 3 - rearranging to $y = mx + c$

Write $2x + y = 7$ in the form $y = mx + c$.

Subtract $2x$ from both sides
 $y = -2x + 7$
 Gradient = -2, y-intercept = 7

$y = -2x + 7$

Rearrange so y is the subject.

Example 4 - finding equation from two points

Find the equation of the line through (0, 3) and (2, 7).

Gradient = $(7 - 3) \div (2 - 0) = 4 \div 2 = 2$
 y-intercept = 3 (the point where $x = 0$)
 $y = 2x + 3$

$y = 2x + 3$

Use two points to find gradient, then identify c .

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying gradient and y-intercept

State m and c for each equation.

- a) $y = 5x + 2$
- b) $y = -3x + 1$
- c) $y = \frac{1}{2}x - 4$
- d) $y = x$

2. Writing equations

Write the equation $y = mx + c$.

- a) Gradient 2, y-intercept 5
- b) Gradient -1, y-intercept 3
- c) Gradient $\frac{3}{4}$, y-intercept -2
- d) Gradient 0, y-intercept 6

3. Rearranging equations

Write in the form $y = mx + c$.

- a) $3x + y = 10$
- b) $2y = 6x - 4$
- c) $x + 2y = 8$
- d) $4x - 2y = 6$

4. From two points

Find the equation of the line through these points.

- a) (0, 1) and (2, 5)
- b) (0, -3) and (3, 6)
- c) (1, 4) and (3, 10)
- d) (2, 1) and (6, 9)

5. Mixed problems

Solve each problem.

- a) Line passes through (0, 4) with gradient -2. Equation?
- b) Which line is steeper: $y = 3x + 1$ or $y = 5x - 2$?
- c) Does (2, 7) lie on $y = 3x + 1$?
- d) Find where $y = 2x - 1$ crosses the x-axis.
- e) Write $6x + 3y = 9$ in $y = mx + c$ form.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) In $y = mx + c$, m is the gradient and c is the y-intercept ☐

Correct answer: _____
Reason: _____

b) $y = 3x + 2$ has y-intercept 3 ☐

Correct answer: _____
Reason: _____

c) A line with gradient 0 is horizontal ☐

Correct answer: _____
Reason: _____

d) $y = 5 - 2x$ has a positive gradient ☐

Correct answer: _____
Reason: _____

e) The y-intercept is found by setting $x = 0$ ☐

Correct answer: _____
Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Line A passes through (1, 5) and (4, 14). Line B passes through (0, 8) and (3, 2). Find the equation of each line and the coordinates of their intersection point.

Expression: _____

Simplified: _____

b) A phone company charges a **monthly fee** plus a **cost per minute**. After 100 minutes the bill is **£18** and after 250 minutes it is **£25.50**. Write the cost C as a function of minutes m , and find the cost for 400 minutes.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

- Have I shown all my working clearly?
- Did I check my answer using a different method?
- Does my answer look reasonable?

